



Techno-Economic Analysis of Distributed Energy Resources (DER)

Integration of Solar PV and Battery Energy Storage Systems

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Abstract

In the context of escalating energy costs and stringent decarbonisation mandates, this study evaluates the techno-economic viability of integrating distributed energy resources (DER) within a commercial facility. Specifically, a 1,002 kWp Solar Photovoltaic (PV) system coupled with a 200 kW / 430 kWh Battery Energy Storage System (BESS) is modelled using a high-fidelity simulation framework. The research employs a Mixed-Integer Linear Programming (MILP) approach to optimize energy dispatch against dynamic wholesale market signals.

Simulation results demonstrate that the proposed configuration generates 1,039,915 kWh annually, theoretically offsetting 149% of the site’s demand. Under an optimized control strategy (Scenario C), the system achieves a self-consumption rate of 26% and yields a net operational cost reduction of approximately 82% compared to the grid-only baseline. Furthermore, the analysis confirms the system’s environmental impact, contributing to an estimated annual carbon abatement of 350 tonnes. These findings underscore the critical role of intelligent dispatch algorithms in maximizing the value of distributed assets in volatile energy markets.

1. Introduction

1.1 Background and Motivation

The global energy landscape is undergoing a paradigm shift, characterized by the rapid penetration of renewable generation and the increasing volatility of electricity markets. For commercial and industrial (C&I) consumers, this transition presents dual challenges: managing rising operational expenditures (OPEX) due to grid tariff instability, and meeting corporate environmental, social, and governance (ESG) targets. Consequently, the adoption of DER, particularly the hybridization of Solar PV and Battery Energy Storage Systems (BESS), has emerged as a strategic imperative.

However, the mere deployment of hardware is insufficient to guarantee economic viability. The stochastic nature of solar generation and the complexity of time-of-use (ToU) or dynamic tariffs require sophisticated energy management strategies to orchestrate the flows between generation, storage, and the grid effectively.

1.2 Study Objectives and Contribution

This paper proposes a rigorous techno-economic framework to assess the feasibility of a bespoke PV-BESS solution for a representative commercial facility. Unlike static sizing methodologies, this study integrates physics-based component modelling with an advanced optimization layer.

The primary contributions of this work are threefold:

1. **High-Fidelity Modelling:** A detailed simulation of PV generation (accounting for thermal and irradiance transposition effects) and non-linear battery throughput.
2. **Scenario Analysis:** A comparative evaluation of three distinct operational modes: a baseline grid-only scenario (A), a standard hardware integration scenario (B), and a fully optimized dispatch scenario utilizing wholesale market signals (C).
3. **Quantified Value Proposition:** A rigorous financial assessment identifying the specific cost-saving mechanisms, including self-consumption maximization and arbitrage.

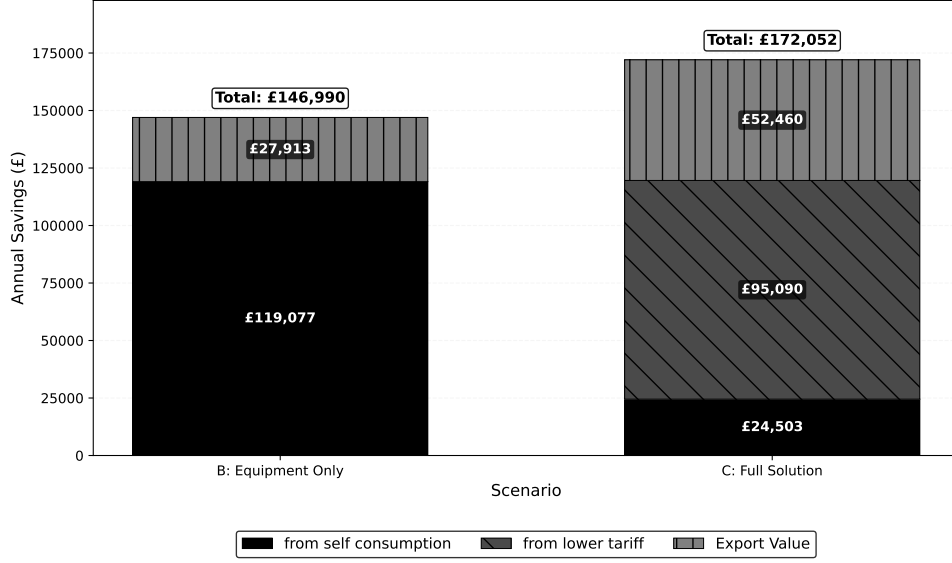


Figure 1. Overview of techno-economic performance: A preview of the annual cost savings composition in Scenario C relative to the baseline, distinguishing the economic contributions of PV self-consumption and BESS arbitrage.

The remainder of this paper is organized as follows: Section 2 details the methodological framework and mathematical formulation. Section 3 presents the simulation results and performance metrics. Section 4 concludes the study with a summary of findings and future outlook.

2. Methodology

The techno-economic feasibility assessment is underpinned by a rigorous simulation framework that integrates high-fidelity physical modelling of renewable assets with advanced mathematical optimization for energy dispatch. This section details the computational methods, mathematical formulations, and parameter assumptions employed.

2.1 Nomenclature

The variable definitions used in the formulation are detailed in Table 1.

Table 1. Nomenclature for Methodology

Symbol	Description	Unit
<i>Sets & Indices</i>		
$\mathcal{T}, t, T$	Set $\mathcal{T} = \{1, \dots, T\}$ with index $t \in \mathcal{T}$	-
<i>Parameters (Vectors)</i>		
$\lambda_{exp/imp}$	Electricity export / import price vector	£/kWh
$\mathbf{S}$	Total Solar Generation Vector	kWh
$\mathbf{L}$	Site Electrical Load Vector	kWh
$\mathbf{A}_{soc}$	Initialization broadcast vector	-
$\mathbf{B}_{ch/dis}$	Charging/Discharging accumulation matrices	h
$\mathbf{1}$	Vector of ones	-
$\mathbf{p}_{norm}$	Normalized typical load profile vector	-
<i>Parameters (Scalars)</i>		
Δt	Time step duration	h
E_{annual}	Total annual electricity consumption	kWh
P_{dc0}	Installed PV DC capacity	kWp
η_{inv}	PV Inverter efficiency	%
L_{sys}	PV System loss factor	%
P_{rate}	Rated AC power of BESS inverter	kW
E_{cap}	Rated energy capacity of BESS	kWh
$\eta_{ch/dis}$	Charging / Discharging efficiency	%
P_{grid}^{max}	Maximum grid connection capacity	kW
SoC_0	Initial State of Charge	kWh
<i>Decision Variables (Vectors $\in \mathbb{R}^T$)</i>		
J	Objective function value (Net Benefit)	£
$\mathbf{P}^{ch/dis}$	Aggregate Charging / Discharging power vectors	kW
$\mathbf{E}^{imp/exp}$	Grid Import / Export energy vectors	kWh
$\mathbf{S}^{load/batt/exp}$	Solar energy flow vectors (to Load, Battery, Grid)	kWh
$\mathbf{S}^{curtail}$	Solar energy curtailed vector	kWh
$\mathbf{G}^{load/batt}$	Grid energy flow vectors (to Load, Battery)	kWh
$\mathbf{D}^{load/exp}$	Battery discharge flow vectors (to Load, Grid)	kWh
$\mathbf{SoC}$	State of Charge vector	kWh

2.2 Framework Overview

The simulation framework operates sequentially to evaluate the techno-economic performance of the distributed energy system, as illustrated in Figure 2. The workflow proceeds as follows:

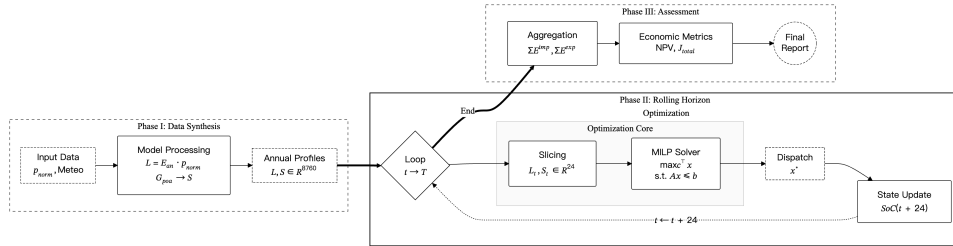


Figure 2. Overview of the Techno-Economic Simulation Framework.

1. Phase 1: Annual Data Synthesis. First, complete annual high-resolution time-series

for electrical load ($\mathbf{L} \in \mathbb{R}^{8760}$) and solar generation ($\mathbf{S} \in \mathbb{R}^{8760}$) are pre-calculated. Load profiles are reconstructed from normalized historical baselines, while PV output is modelled using physics-based irradiance transposition and thermal algorithms.

2. **Phase 2: Rolling Horizon Optimization.** With the full annual datasets established, the Energy Management System (EMS) performs a daily rolling simulation. For each day of the year, the optimizer extracts the data corresponding to the 24-hour prediction horizon $\mathcal{T}$ and employs a MILP algorithm to derive the optimal dispatch strategy for that specific day.
3. **Phase 3: Economic Assessment.** By coupling physical asset constraints with real-time market tariffs, the solver determines the global optimum maximizing the net economic benefit, thereby quantifying the theoretical performance of the system.

2.3 Load Profiling and Decomposition

To construct a representative simulation of the site’s energy dynamics, the annual aggregate energy consumption is decomposed into a granular hourly load profile. Let E_{annual} denote the total annual electricity consumption (in kWh). We introduce a normalized typical load curve vector $\mathbf{p}_{norm} \in \mathbb{R}^{8760}$, representing the per-unit power demand profile derived from historical data.

The normalization constraint ensures the profile sums to unity:

$$\mathbf{1}^\top \mathbf{p}_{norm} = 1 \quad (1)$$

The specific hourly load vector $\mathbf{L} \in \mathbb{R}^{8760}$ for the site is synthesized by scaling this normalized profile:

$$\mathbf{L} = E_{annual} \times \mathbf{p}_{norm} \quad (2)$$

This synthesized annual load vector $\mathbf{L}$ serves as a primary input for the subsequent rolling optimization, where it is segmented into daily forecast horizons.

2.4 Solar Photovoltaic Generation Modelling

The solar generation profile is modelled using the `pvlb` python library [2]. The physics-based modelling chain proceeds through three distinct stages:

2.4.1 Irradiance Transposition

First, the Global Horizontal Irradiance (GHI) and Diffuse Horizontal Irradiance (DHI) are converted to Plane-of-Array (POA) irradiance (G_{poa}) using the Hay-Davies transposition model

[3], which accounts for the anisotropic nature of diffuse radiation:

$$G_{poa} = G_{beam}R_b + DHI \left(A_i R_b + (1 - A_i) \frac{1 + \cos \beta}{2} \right) + GHI \rho_g \frac{1 - \cos \beta}{2} \quad (3)$$

Where R_b is the geometric factor for beam radiation (dependent on the module tilt β and azimuth γ), A_i is the anisotropy index, and ρ_g is the ground albedo.

2.4.2 Thermal Modelling

Second, the Cell temperature is estimated using the Sandia Array Performance Model (SAPM) [4], which dynamically adjusts for ambient temperature (T_{air}), wind speed (v_{wind}), and incident irradiance:

$$T_{cell} = T_{air} + \exp(a + b \cdot v_{wind}) \cdot G_{poa} + \Delta T \quad (4)$$

2.4.3 Power Conversion

DC power output is calculated using the PVWatts semi-empirical model [5], explicitly accounting for the temperature coefficient of power (γ_{pdc}):

$$P_{dc} = \frac{G_{poa}}{1000} \cdot P_{dc0} \cdot [1 + \gamma_{pdc}(T_{cell} - T_{ref})] \quad (5)$$

Finally, the AC grid injection power S_t at each time step is derived by applying the inverter efficiency (η_{inv}) and the comprehensive system loss factor (L_{sys}):

$$S_t = P_{dc,t} \cdot \eta_{inv} \cdot (1 - L_{sys}) \quad (6)$$

The resulting vector $\mathbf{S} \in \mathbb{R}^{8760}$, synchronized with the load profile, completes the boundary conditions required for the EMS dispatch logic.

2.5 Optimization Algorithm and Control Strategy

The dispatch strategy for the BESS is governed by a centralized EMS. We formulate the energy management problem as a MILP model over a rolling prediction horizon $\mathcal{T} = \{1, \dots, T\}$, where T typically represents a 24-hour day-ahead window. The solver identifies the global optimum for power flows to maximize the objective function.

2.5.1 Mathematical Formulation

The complete optimization problem is defined by the objective function (8) subject to the physical and operational constraints (9)–(19). Let $T = |\mathcal{T}|$ denote the optimization horizon length. We define the decision variable vector $\mathbf{x} \in \mathbb{R}^n$ containing all decision variables over the horizon, where n is the total number of decision variables.

Decision Variables: The decision vector $\mathbf{x}$ is composed of energy flow variables concatenated over all time steps:

$$\mathbf{x} = [\mathbf{S}^{load}, \mathbf{S}^{exp}, \mathbf{S}^{bess}, \mathbf{S}^{curtail}, \mathbf{D}^{load}, \mathbf{D}^{exp}, \mathbf{G}^{load}, \mathbf{G}^{batt}, \mathbf{E}^{imp}, \mathbf{E}^{exp}, \mathbf{P}^{ch}, \mathbf{P}^{dis}, \mathbf{SoC}]^T \quad (7)$$

Objective Function

The objective is to maximize the objective function J , representing the net economic benefit, calculated as the scalar product of price vectors and energy flow vectors:

$$\max J = \underbrace{\boldsymbol{\lambda}_{exp}^\top \mathbf{E}^{exp}}_{\text{I}} - \underbrace{\boldsymbol{\lambda}_{imp}^\top \mathbf{E}^{imp}}_{\text{II}} + \underbrace{\boldsymbol{\lambda}_{imp}^\top \mathbf{S}^{load}}_{\text{III}} \quad (8)$$

Where:

- **I (Export Revenue):** Revenue from grid exports.
- **II (Import Cost):** Cost of grid imports.
- **III (Self-Consumption Value):** Avoided cost from using solar energy directly.

Constraints

$$\mathbf{S}^{load} + \mathbf{S}^{exp} + \mathbf{S}^{bess} + \mathbf{S}^{curtail} = \mathbf{S} \quad (9)$$

$$\mathbf{S}^{load} + \mathbf{D}^{load} + \mathbf{G}^{load} = \mathbf{L} \quad (10)$$

$$\mathbf{E}^{imp} = \mathbf{G}^{load} + \mathbf{G}^{batt} \quad (11)$$

$$\mathbf{E}^{exp} = \mathbf{S}^{exp} + \mathbf{D}^{exp} \quad (12)$$

$$\mathbf{P}^{ch} = \mathbf{S}^{bess} + \mathbf{G}^{batt} \quad (13)$$

$$\mathbf{P}^{dis} = \mathbf{D}^{load} + \mathbf{D}^{exp} \quad (14)$$

$$\mathbf{SoC} = \mathbf{A}_{soc} \mathbf{SoC}_0 + \mathbf{B}_{ch} \mathbf{P}^{ch} - \mathbf{B}_{dis} \mathbf{P}^{dis} \quad (15)$$

$$\mathbf{0} \leq \mathbf{SoC} \leq E_{cap} \mathbf{1} \quad (16)$$

$$\mathbf{0} \leq \mathbf{P}^{ch} \leq P_{rate} \mathbf{1} \quad (17)$$

$$\mathbf{0} \leq \mathbf{P}^{dis} \leq P_{rate} \mathbf{1} \quad (18)$$

$$\mathbf{E}^{imp}, \mathbf{E}^{exp} \leq P_{grid}^{max} \Delta t \mathbf{1} \quad (19)$$

The complete constraint set can be expressed in compact matrix form as:

$$\mathbf{A}_{eq} \mathbf{x} = \mathbf{b}_{eq} \quad (20)$$

$$\mathbf{A}_{ineq} \mathbf{x} \leq \mathbf{b}_{ineq}, \quad \mathbf{x} \geq \mathbf{0} \quad (21)$$

2.5.2 Constraint Interpretation

Interpretation of SoC Dynamics (Eq. 15) To reformulate the recursive State-of-Charge (SoC) difference equation into a simultaneous linear system suitable for the solver, we express the SoC at any time step t as the initial state plus the cumulative sum of all past charging and discharging events. This integration process is encoded directly into the lower-triangular accumulation matrices $\mathbf{B}_{ch}$ and $\mathbf{B}_{dis}$, which also embed the round-trip efficiency losses:

- $\mathbf{A}_{soc}$ is an initialization vector ($\mathbf{1} \in \mathbb{R}^{T \times 1}$) that broadcasts the initial state $\mathbf{SoC}_0$ across the horizon.
- $\mathbf{B}_{ch}$ is the **Charging Accumulation Matrix** ($\in \mathbb{R}^{T \times T}$), a lower triangular matrix scaled by the charging efficiency η_{ch} . It sums up all energy charged up to time t :

$$\mathbf{B}_{ch} = \Delta t \cdot \eta_{ch} \cdot \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}$$

- $\mathbf{B}_{dis}$ is the **Discharging Accumulation Matrix** ($\in \mathbb{R}^{T \times T}$), scaled by the inverse of discharging efficiency $1/\eta_{dis}$. This accounts for the fact that to discharge 1 unit of energy

to the grid, more than 1 unit must be drained from the battery:

$$\mathbf{B}_{dis} = \frac{\Delta t}{\eta_{dis}} \cdot \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}$$

Other Constraints The remaining constraints ensure physical feasibility and adherence to operational limits:

- **Solar Energy Balance** Eq. (9): Ensures that for the entire horizon, the generated solar energy vector $\mathbf{S}$ is exactly balanced by the sum of self-consumption, storage, export, and curtailment vectors.
- **Load Satisfaction** Eq. (10): Guarantees that the site load vector $\mathbf{L}$ is fully met by a linear combination of direct solar, battery discharge, and grid import vectors.
- **Grid Exchange Definitions** Eq. (11) & (12): Define the total grid import $\mathbf{E}^{imp}$ and export $\mathbf{E}^{exp}$ vectors based on their constituent flow components.
- **Inverter Power Definitions** Eq. (13) & (14): Aggregate power flows to define the total charging $\mathbf{P}^{ch}$ and discharging $\mathbf{P}^{dis}$ power vectors at the battery inverter terminals.
- **Capacity Limits** Eq. (16): Constrains the State of Charge vector $\mathbf{SoC}$ to remain within $[0, E_{cap}]$ for all time steps.
- **Power Ratings** Eq. (17) & (18): Ensures that the charging and discharging power vectors do not exceed the rated power capacity P_{rate} of the Power Conversion System (PCS).
- **Grid Connection Limits** Eq. (19): Ensures that the total power exchanged with the grid (both import and export vectors) respects the site's maximum authorized grid connection capacity P_{grid}^{max} .

2.6 Assumptions and Limitations

2.6.1 System Topology and Design

The simulation is predicated on a specific system configuration designed to balance technical constraints with economic viability. The proposed topology incorporates a 1,002 kWp PV array, sized to fully utilize the available 5,000 m² roof area. The 200 kW / 430 kWh BESS is dimensioned to align with the 200 kW PCS and the existing 200 kW grid connection, thereby avoiding costly infrastructure reinforcement.

Figure 3 illustrates the electrical single-line diagram (SLD), defining the nodal relationships between the PV generation, battery storage, site load, and the grid point of common coupling (PCC).

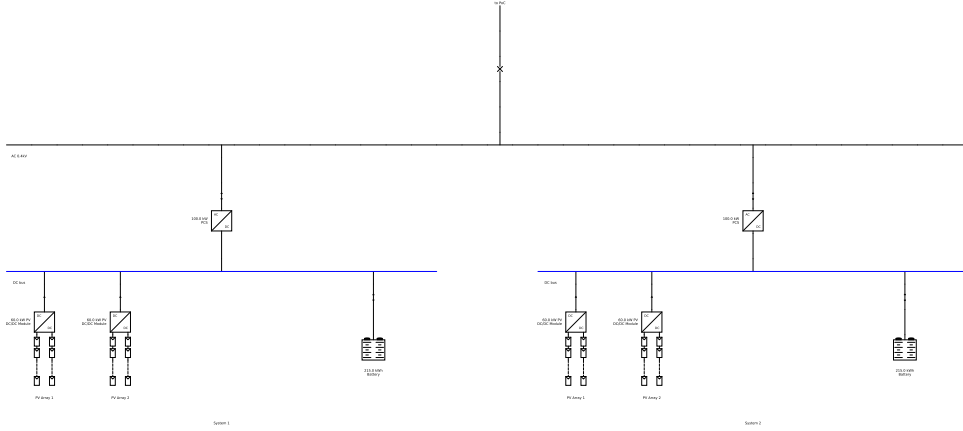


Figure 3. SLD of the Proposed System

The system implementation utilizes industry-standard components that align with the defined technical specifications. The key hardware parameters for the proposed configuration are summarized in Table 2.

Table 2. Core Equipment List

Category	Model	Key Specs	Qty
PV Panel	BYD NLBK-27-435W	435 W	2,304
BESS	EnerArk-NBN-P100	215 kWh	2
PCS	EnerP CU	100 kW	2

2.6.2 Modelling Parameters

Table 3 consolidates the key parameters used in the modelling.

Table 3. Summary of Modelling Parameters and Assumptions

Category	Parameter (Symbol)	Value / Assumption
<i>1. Load Profiling</i>		
Site Data	Site Postcode	SW1A 1AA
	Annual Demand (E_{annual})	700,000 kWh
Grid	Grid Import Limit (P_{limit}^{imp})	200 kW
	Grid Export Limit (P_{limit}^{exp})	266 kW
Economics	Peak Import Tariff (λ_{peak})	£0.22 / kWh
	Valley Import Tariff (λ_{valley})	£0.18 / kWh
<i>2. Solar PV Generation</i>		
Configuration	Available Roof Area	5,000 m^2
	Installed Capacity (P_{dc0})	1,002 kWp
	Module Tilt Angle (β)	35°
	Module Azimuth Angle (γ)	180° (South=180)
Modelling	Meteo Data Source	PVGIS TMY [6]
	Transposition Model	Hay-Davies [3]
	Thermal Model	SAPM (Open Rack) [4]
Performance	Temp. Coefficient (γ_{pdc})	-0.4% / °C
	System Loss Factor (L_{sys})	14%
	Reference Temp. (T_{ref})	25°C
<i>3. BESS Optimization</i>		
Configuration	Rated Power (P_{rate})	200 kW
	Energy Capacity (E_{cap})	430 kWh
Performance	Round-Trip Efficiency (η_{rte})	92%
	One-way Efficiency ($\eta_{ch/dis}$)	$\sqrt{\eta_{rte}}$
	Initial SoC (SoC_0)	0% E_{cap}
Algorithm	Method	MILP
	Solver	GLPK_MI
	Time Resolution (Δt)	1 Hour
	Forecast Horizon (T)	24 Hours

2.6.3 Limitations

The methodology presented herein is subject to several intrinsic limitations that define the boundaries of the simulation’s applicability:

- 1. Load Profile Approximation:** The annual load profile is synthesized from a normalized typical curve ($\mathbf{p}_{norm}$) derived from an internal database, rather than high-resolution historical metering data from the specific site. This approach assumes standard operational patterns and may not capture site-specific anomalies or unique demand fluctuations.
- 2. Deterministic Foresight:** The rolling optimization in Phase 2 operates under the assumption of perfect foresight within each prediction horizon T . The MILP model utilizes deterministic knowledge of future load, solar generation, and market prices. In operational

reality, stochastic forecast errors would inevitably lead to deviations from this theoretically optimal dispatch schedule.

3. **Linearized Battery Physics:** To maintain computational tractability, the BESS is modelled using a linear formulation with constant charging/discharging efficiencies (η_{ch}, η_{dis}). This approach neglects non-linear electrochemical behaviors such as C-rate dependent efficiency (Peukert’s law), temperature sensitivity of capacity, and long-term state-of-health (SoH) degradation due to cycle aging.
4. **Temporal Granularity:** The simulation employs a one-hour temporal resolution ($\Delta t = 1$ h). While sufficient for arbitrage-driven dispatch, this granularity effectively smooths out sub-hourly power fluctuations. Consequently, the model may underestimate instantaneous peak power demands or the potential value from high-frequency ancillary services (e.g., frequency response).

3. Results

This study evaluates three distinct operational scenarios to quantify the techno-economic performance of the proposed distributed energy system. The scenario definitions are as follows:

Scenario A (Baseline): Represents the reference case where the site relies exclusively on grid imports to meet electrical demand, with no on-site generation or storage assets.

Scenario B (Retail Optimization): Integrates the proposed PV and BESS capacity. The EMS optimizes dispatch to minimize costs based on the standard retail tariff structure (e.g., Time-of-Use), prioritizing self-consumption and peak shaving.

Scenario C (Wholesale Optimization): Utilizes the identical hardware configuration as Scenario B. However, the EMS is configured to optimize dispatch against dynamic wholesale market price signals, aiming to maximize value through arbitrage and opportunistic grid interaction.

Table 4. Simulation Scenario Definitions

Scenario	PV Capacity	BESS Capacity	Price
A	0 kWp	0 kWh	Retail Tariff
B	1,002 kWp	430 kWh	Retail Tariff
C	1,002 kWp	430 kWh	Wholesale Market

3.1 System Characterization and Market Dynamics

3.1.1 Wholesale Market Price Dynamics

The economic performance of the BESS is intrinsically linked to the tariff structure governing grid interactions. Figure 4 contrasts the "bundled" retail tariff used in Scenario B with the dynamic wholesale pricing mechanism of Scenario C. The retail tariff imposes a smoothed cost structure (typically flat or simple Time-of-Use), which effectively masks the underlying market signals. While this provides budget certainty, it severely limits the BESS's ability to generate value, constraining it to simple peak-shaving operations.

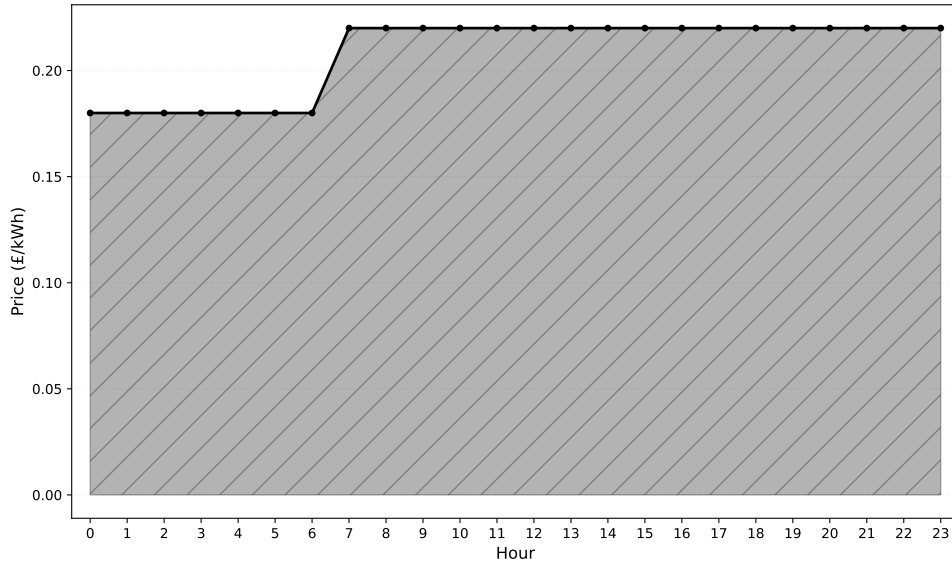


Figure 4. Static Retail Tariff Structure

In contrast, Scenario C exposes the system to the unbundled cost vector λ_t^{imp} , defined as:

$$\lambda_t^{imp} = P_{wholesale,t} + P_{network,t} + P_{fixed} \quad (22)$$

Where $P_{wholesale,t}$ is the spot price and $P_{network,t}$ represents volatile grid charges (TNUoS/GDUoS). P_{fixed} comprises non-volatile levies. As visualized in Figure 5, this dynamic pricing reflects real-time supply-demand balances, exhibiting distinct temporal patterns that the MILP algorithm leverages for scheduling.

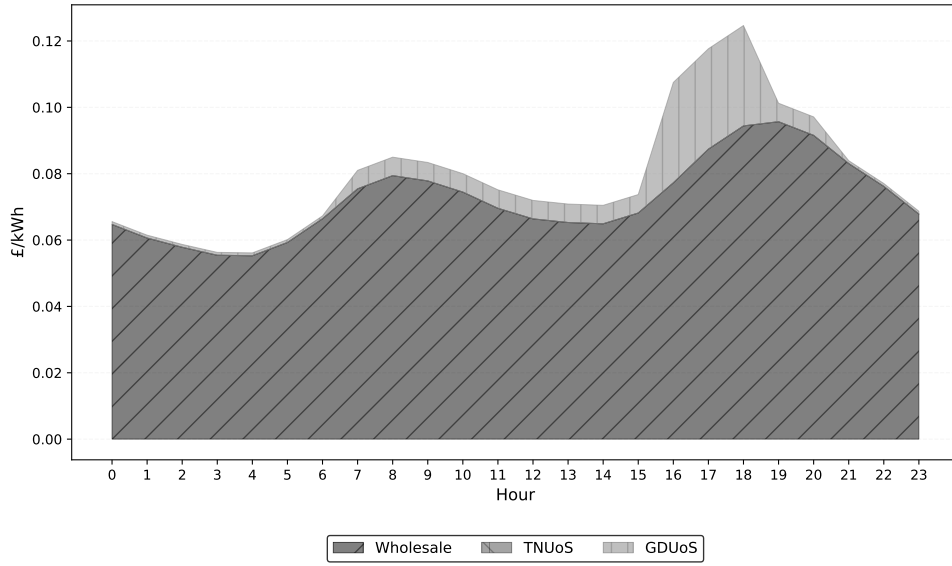


Figure 5. Temporal Evolution of Wholesale Market Costs. Data source: Nord Pool, FS Energy [7] [8].

Critically, it is the magnitude of the price spread—rather than the average price—that drives arbitrage revenue. Figure 6 analyzes this statistical dispersion. The significant variance between the minimum (often negative) and maximum prices creates the arbitrage spread. The EMS in Scenario C exploits these extremes: charging during low-cost intervals (lower quartiles) and discharging during scarcity events (upper quartiles), thereby capturing value that is inaccessible under the rigid retail tariff structure of Scenario B.

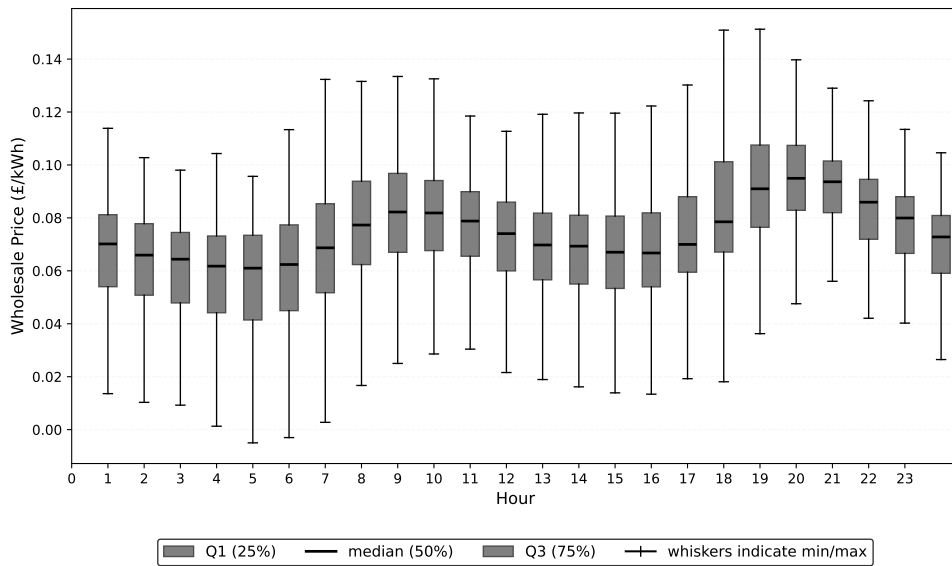


Figure 6. Statistical Distribution of Wholesale Market Volatility. Data source: Nord Pool, FS Energy.

3.1.2 Load and Generation Characteristics

Table 5 outlines the site’s annual energy consumption relative to the potential PV generation. The analysis indicates that the proposed PV system capacity is sufficient to theoretically cover nearly 149% of the total annual consumption.

Table 5. Annual Load and PV Generation Comparison

Category	Annual Energy
Total Site Load	700,000 kWh
Proposed PV Generation	1,039,915 kWh

Figure 7 depicts the monthly energy profile, revealing a distinct temporal mismatch between load demand and solar generation. Peak PV yield coincides with high-irradiance summer months, creating seasonal surpluses. Conversely, winter months exhibit deficits. This seasonality necessitates a dynamic dispatch strategy to maximize self-consumption during surplus periods and leverage energy arbitrage during deficit periods.

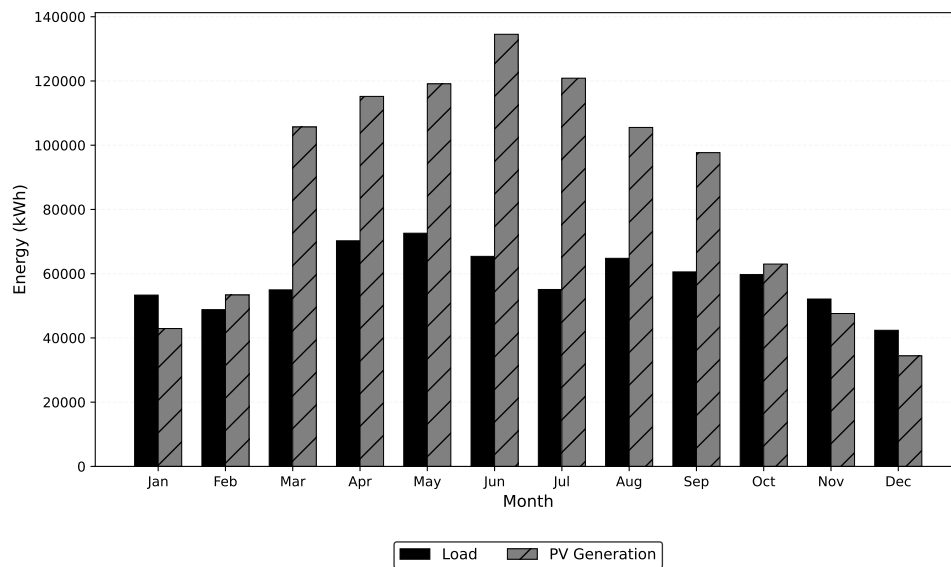


Figure 7. Monthly Load Characteristics and PV Generation

Figure 8 presents the aggregated 24-hour diurnal load profile. The profile exhibits a consistent generation surplus during peak irradiance hours (10:00–14:00). The BESS is essential to capture this surplus and mitigate curtailment. During evening peak hours (17:00–21:00), the BESS discharges stored energy to displace high-cost grid imports, effectively flattening the net load profile.

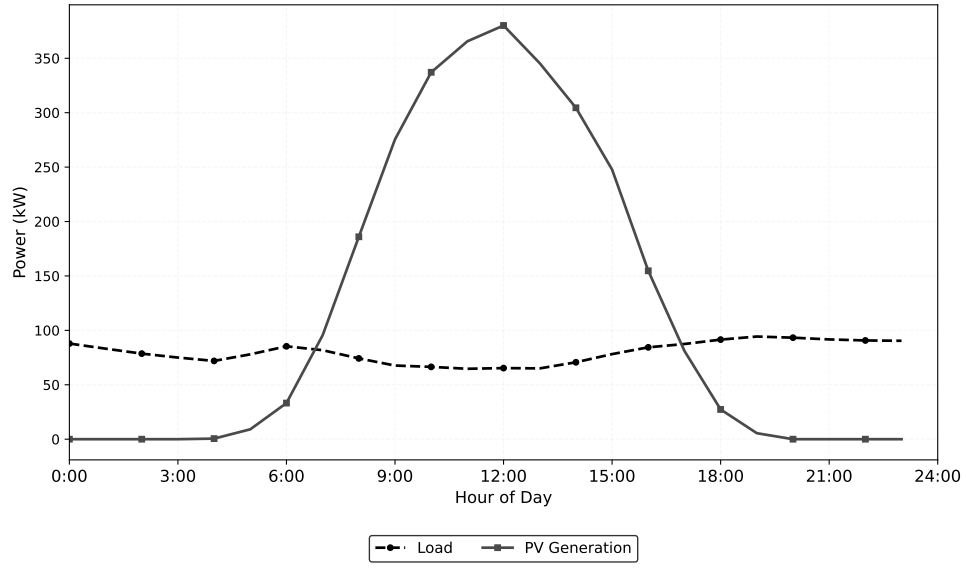


Figure 8. Diurnal Load Profile and PV Generation

3.2 Operational Dynamics Analysis

3.2.1 Energy Flow Analysis

Figure 9 delineates the energy flow dynamics for the active dispatch scenarios (B and C), quantifying the source-to-load transitions.

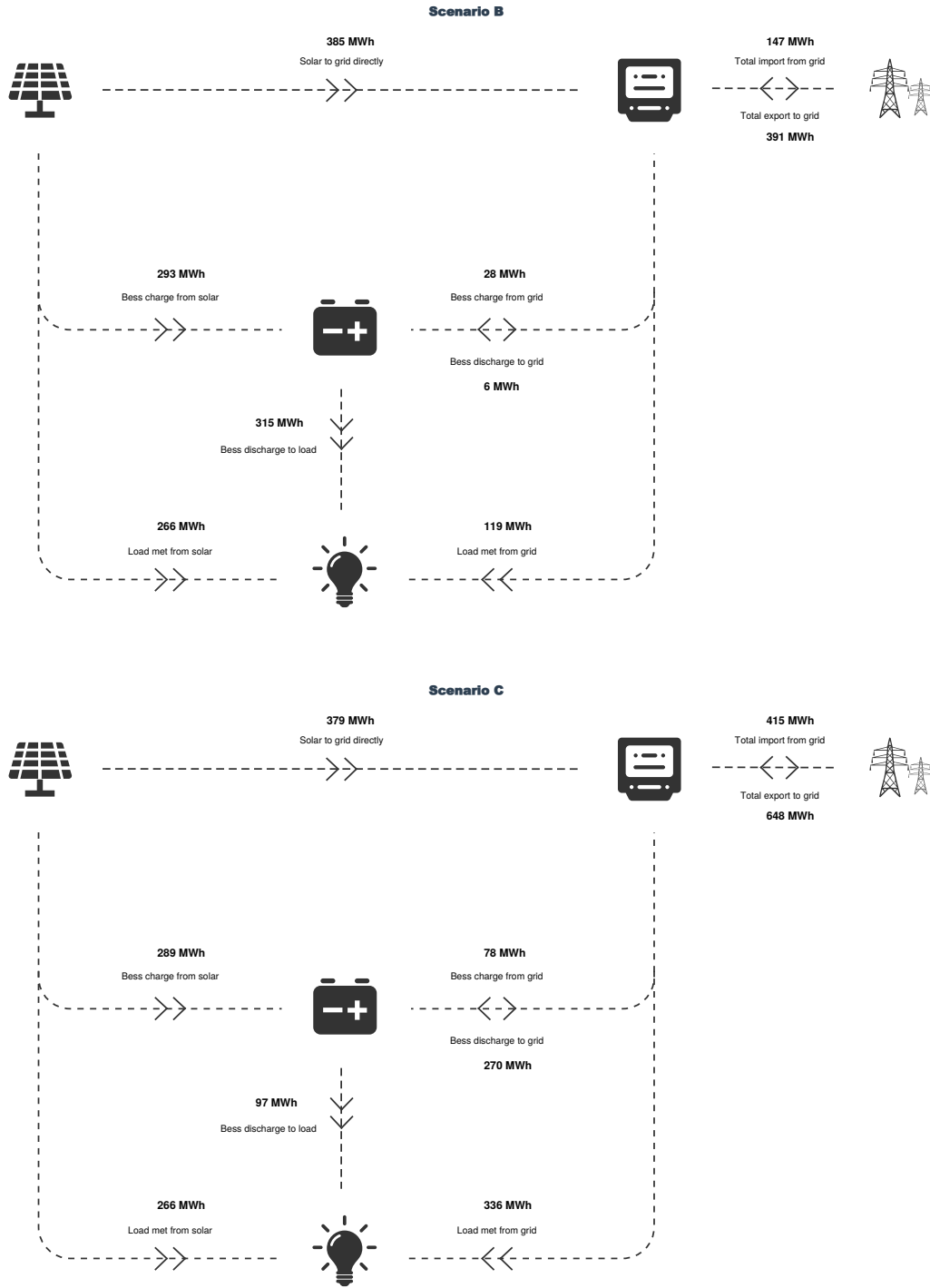


Figure 9. Energy Flow Pathways and Source Contributions.

Scenario A functions as a passive load, meeting its total annual demand of 700 MWh solely through grid importation.

In Scenario B, the system incorporates local generation and storage, with the EMS optimization driven by the retail ToU tariff structure. The PV system contributes 1,040 MWh, reducing grid imports to 147 MWh. The BESS operation (642 MWh) is determined by minimizing the cost function against the predictable peak/off-peak retail price vector, effectively prioritizing

self-consumption and peak load reduction.

Scenario C utilizes the identical hardware and MILP formulation but replaces the retail tariff with the dynamic wholesale market price vector in the objective function. While the total PV yield is invariant, the BESS dispatch profile (734 MWh) is reconfigured by the solver to exploit market volatility. The control logic thus shifts from tariff-based avoidance to dynamic price arbitrage, scheduling power flows to maximize the net economic benefit defined in Eq. (8).

3.3 Techno-Economic Performance Evaluation

3.3.1 Economic Performance Analysis

The primary metric for evaluating system viability is the reduction in annual OPEX. First, the annual electricity cost C for each scenario is derived by aggregating the gross expenditure on grid energy imports:

$$C = \sum_{t \in \mathcal{T}} \lambda_t^{imp} E_t^{imp} \quad (23)$$

Where λ_t^{imp} denotes the time-dependent import tariff, and E_t^{imp} represents the total grid import volume at time step t (comprising both direct load supply and battery charging).

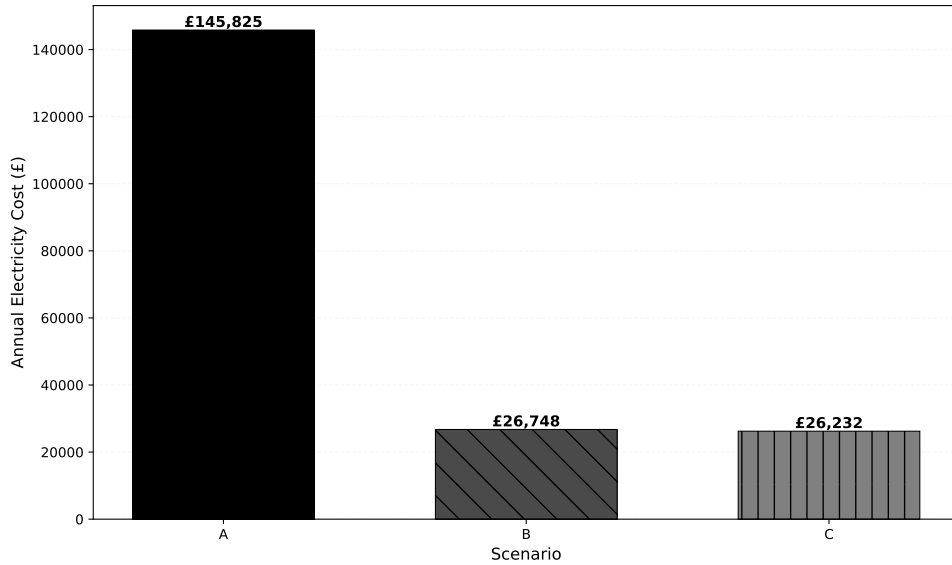


Figure 10. Annual Electricity Cost Comparison by Scenario

Figure 10 visually decomposes the cost structures. While Scenario B achieves savings primarily through avoided grid imports (self-consumption), Scenario C further compresses the net expenditure by strategically time-shifting grid interactions—minimizing import costs during price valleys and generating revenue during price peaks.

3.3.2 System Efficiency Indicators

To rigorously quantify the system's technical efficiency, we define three key performance metrics over the simulation horizon $\mathcal{T}$: PV Utilization, Self-Sufficiency Ratio (SSR), and Grid Dependency. Let S_t denote PV generation, L_t load demand, P_t^{ch} battery charging power, and E_t^{imp} grid imports at time t .

First, **PV Utilization** measures the proportion of generated solar energy that is consumed on-site (either directly or via storage), reflecting the effectiveness of the system in minimizing export/curtailment:

$$\text{PV Utilization} = \frac{\sum_{t \in \mathcal{T}} \min(S_t, L_t + P_t^{ch})}{\sum_{t \in \mathcal{T}} S_t} \quad (24)$$

Second, the **Self-Sufficiency Ratio (SSR)** quantifies the extent to which the site's load is met by on-site resources (solar and battery discharge):

$$\text{SSR} = \frac{\sum_{t \in \mathcal{T}} (S_t^{load} + D_t^{load})}{\sum_{t \in \mathcal{T}} L_t} \quad (25)$$

Finally, **Grid Independence** measures the proportion of total system consumption (load plus grid-charging) satisfied by local generation:

$$\text{Grid Independence} = \frac{\sum_{t \in \mathcal{T}} (S_t^{load} + D_t^{load})}{\sum_{t \in \mathcal{T}} (L_t + G_t^{batt})} \quad (26)$$

As shown in Figure 11, these metrics are compared across scenarios to evaluate the technical impact of the proposed system.

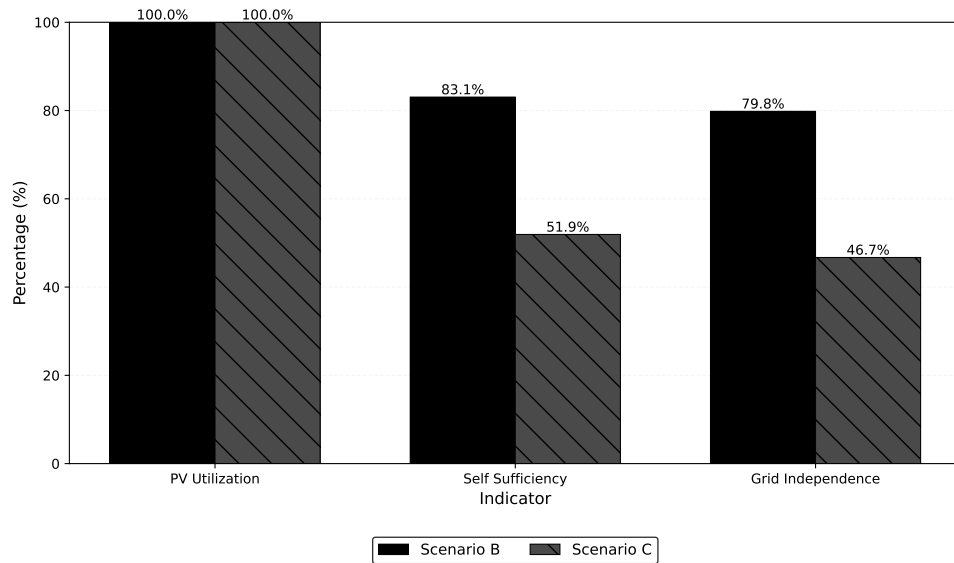


Figure 11. Comparison of Energy Performance Indicators

The integration of PV and BESS assets (Scenarios B and C) induces a paradigm shift in the site’s operational profile, transitioning from absolute grid reliance (Scenario A) to a regime of partial autonomy. A critical observation, however, is the divergence in performance drivers between Scenario B and Scenario C. While Scenario B maximizes self-sufficiency through rule-based import avoidance, Scenario C exposes the nuanced trade-off governed by the MILP objective function. The results indicate that the solver strategically subordinates pure energetic autonomy to financial efficiency, sacrificing marginal self-sufficiency to capitalize on high-value arbitrage windows. This validates the hypothesis that in a volatile market environment, the optimal dispatch strategy is defined not by physical isolation, but by the maximization of economic utility.

3.4 Conclusion of Results

The quantitative analysis confirms that the proposed PV-BESS system, when coupled with an optimized dispatch strategy, delivers substantial techno-economic benefits. The transition from the baseline (Scenario A) to the hardware-integrated Scenario B effectively addresses energy security and self-sufficiency. However, it is the algorithmic intelligence of Scenario C that unlocks the system’s full financial potential. By dynamically responding to wholesale market signals, the MILP framework achieves a robust operational cost reduction of approximately 82%, validating that economic efficiency in volatile markets requires active, forecast-driven management rather than passive rule-based control.

4. Conclusion

This study developed and validated a holistic techno-economic framework for distributed energy resources, integrating physics-based asset modelling with advanced mathematical optimization. The research demonstrates that for commercial prosumers, the value of renewable assets is no longer defined solely by installed capacity, but by the intelligence of the dispatch strategy.

The findings establish that while hardware integration provides the necessary physical flexibility, it is the optimization layer that monetizes this flexibility against market volatility. The superior performance of the proposed solution underscores the critical role of MILP in orchestrating complex trade-offs between self-consumption and arbitrage. Ultimately, this work provides a scalable blueprint for transforming passive energy consumers into active, economically optimized market participants.

5. Limitations and Future Work

The conclusions drawn from this study represent a theoretical upper bound on system performance, subject to the practical constraints of real-world deployment. While the modeling

limitations regarding forecast uncertainty and battery physics have been established in the Methodology, two broader implementation challenges warrant specific attention. First, the financial feasibility is highly sensitive to the regulatory environment; specifically, Distribution Network Operator (DNO) constraints on export capacity could materially impact the projected revenue streams. Second, the structural suitability of the site and the long-term stability of the assumed market tariff structures introduce external variables that extend beyond the scope of the current techno-economic model.

Future research will aim to bridge the gap between theoretical optimization and operational reality. A primary focus will be the integration of stochastic programming or Model Predictive Control (MPC) to address the forecast uncertainties inherent in real-time dispatch. Additionally, subsequent research will refine the asset modeling to include non-linear degradation costs and expand the boundary conditions to include ancillary service markets (e.g., Frequency Response) for additional revenue stacking.

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